

Statement of Work

Project Title

Crime Data Engineering Pipeline and Reporting Dataset

Project Background

This project was completed as part of a Python Data Engineering assignment focused on preparing operational UK police crime data for downstream Business Intelligence reporting and analysis.

The project simulates the role of a Data Engineer working within a Police Force Analytics Unit, responsible for building scalable, reusable, and validated data pipelines capable of supporting operational decision-making.

Project Objective

The objective of the project was to design and implement a complete Python ETL (Extract, Transform, Load) pipeline that:

- ingests large-scale UK police crime datasets
- cleans and validates operational data
- enriches crime data using external contextual datasets
- aggregates data to a consistent reporting grain
- exports a final BI-ready reporting dataset

The final output was designed for use within Power BI dashboards and operational reporting workflows.

Datasets Used

Primary Dataset:

UK Police Crime Data:

- Source: <https://data.police.uk/data/>

- Four police forces selected:
 - Metropolitan Police
 - Merseyside Police
 - West Midlands Police
 - Northumbria Police

Enrichment Datasets:

1. Football Match Data

Source:

<https://www.football-data.co.uk/>

Purpose:

Football match activity was incorporated to investigate whether football events and derby fixtures could be associated with changes in crime levels.

Derived measures:

- home_match_count
- derby_match_count

2. Population Data

Source:

Office for National Statistics (ONS)

Purpose:

Population estimates were integrated to create normalised crime-rate metrics and enable fair comparisons between police-force regions.

Derived measure:

- crime_rate_per_1000

Scope of Work

The project included the following stages:

1.Data Ingestion

- Importing multiple monthly crime CSV files
- Importing football season datasets
- Importing ONS population data
- Processing files iteratively to improve scalability

2.Data Cleaning and Validation

- Standardising column names
- Cleaning date fields
- Handling missing values
- Duplicate validation
- Row-count reconciliation
- Validation at reporting grain

3.Feature Engineering and Transformation

- Creating year and month fields
- Creating football match indicators
- Creating derby-match indicators
- Aggregating crime counts
- Calculating crime-rate KPIs

4. Data Aggregation

The final reporting dataset was aggregated to:

Police Force × Month × Crime Type

This aggregation level was selected to support:

- operational reporting
- trend analysis
- force comparison

- downstream BI dashboarding

5. Data Enrichment

The project integrated:

- football match data
- population estimates

Joins were validated to prevent unintended many-to-many duplication.

6. Exporting and Reporting Readiness

The final output was exported as a clean CSV dataset suitable for:

- Power BI dashboards
- KPI monitoring
- trend analysis
- operational reporting

Assumptions

- Annual population estimates were applied across all months within each corresponding year.
- 2024 population estimates were reused for 2025 due to limited subnational availability.
- Football enrichment focused on selected Premier League clubs associated with chosen police-force regions.
- Dropna() was used in the football match dataset as data without home teams, away teams, or a date are completely useless and there is no way to combat it.

Deliverables:

The project deliverables include:

- Python ETL pipeline notebook
- Final reporting-ready CSV dataset
- Workflow diagram
- README documentation
- Data dictionary
- Validation outputs
- Statement of Work

Final Output

The final output dataset contains:

- aggregated crime metrics
- football enrichment measures
- population estimates
- normalised crime rates

and is designed for downstream Power BI analysis and operational reporting.